

Efficient Packaging and Deployment of AI Models for Edge Inference

A Reproducible MobileNetV2 FP32-to-INT8 Study

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Abstract—Edge inference systems require a compact model and a reproducible path from conversion to serving. This paper presents an open MobileNetV2 deployment artifact that generates comparable FP32 and full-integer TensorFlow Lite models, evaluates both on the complete 3,925-image Imagenette validation split, measures single-thread host-CPU latency, and serves the INT8 artifact through FastAPI and Docker. Calibration uses 200 deterministically selected training images. On an AMD Ryzen 5 7600 with TensorFlow 2.15.1 and one TFLite thread, integer quantization reduced the model from 13.34 to 3.81 MiB (3.50 $\times$), reduced mean inference latency from 8.53 to 6.46 ms (1.32 $\times$), and changed top-1 accuracy from 66.73% to 65.81% (a 0.92 percentage-point decrease). The artifact records raw latency samples, hashes, preprocessing, versions, and dataset provenance. These host results establish the conversion pipeline; device-specific latency, energy, and container overhead remain separate measurements for target edge hardware.

Index Terms—edge inference, integer quantization, MobileNetV2, TensorFlow Lite, reproducibility, containerization

I. INTRODUCTION

Executing inference near the data source can reduce network dependence and response time, but edge devices impose tight limits on storage, memory, computation, and power. Deployment therefore includes both model transformation and software packaging. A small artifact that cannot be reproduced, loaded, or monitored is not a complete edge system.

Integer post-training quantization maps floating-point weights and activations to eight-bit representations and can reduce model size while enabling integer kernels [1], [2]. Compact architectures such as MobileNetV2 complement this transformation through depthwise separable convolutions and inverted residual blocks [3]. Prior benchmark efforts have also shown the need to report latency, memory, and accuracy under explicit hardware and software conditions [4].

This work contributes a public, executable artifact rather than a new quantization algorithm. The artifact (1) downloads a checksum-verified evaluation dataset, (2) converts one pretrained graph to FP32 and full-integer TFLite formats, (3) evaluates both formats with identical preprocessing and labels, (4) records raw timing samples and provenance, and (5) exposes the optimized model through a bounded inference API and container. The central question is practical: what changes when precision is reduced while the architecture,

dataset, preprocessing, runtime family, and thread count are held fixed?

II. METHODOLOGY

A. Model conversion

The source model is MobileNetV2 with ImageNet weights and a $224 \times 224 \times 3$ input. TensorFlow 2.15.1 exports the same SavedModel graph twice. The first conversion retains FP32 tensors. The second enables default optimization, restricts operators to the built-in INT8 set, and requests signed INT8 inputs and outputs. A representative dataset of 200 training images calibrates activation ranges. For scale S , zero point Z , and integer bounds $q_{\min}$ and $q_{\max}$, an input value x is mapped as

$$q = \text{clip}(\text{round}(x/S) + Z, q_{\min}, q_{\max}). \quad (1)$$

Both conversion outputs are retained with SHA-256 hashes.

Pruning is outside the reported pipeline. No pruning accuracy or speed claim is made.

B. Dataset and accuracy protocol

Imagenette is a ten-class subset of ImageNet containing tench, English springer, cassette player, chain saw, church, French horn, garbage truck, gas pump, golf ball, and parachute [5]. The 160-pixel distribution contains 9,470 training and 3,925 validation images. Its archive is identified by SHA-256 digest 64d0c485...db02fe5. Calibration images are selected from the training split with seed 20260914. All validation images are evaluated.

Each image is converted to RGB, resized so its shortest side is 256 pixels, center-cropped to 224×224 , and scaled to $[-1, 1]$. Predicted indices are compared with the corresponding ImageNet class indices. We report top-1 and top-5 accuracy and Wilson 95% confidence intervals. The use of ImageNet-pretrained weights means this experiment evaluates conversion fidelity on an ImageNet subset; it is not an Imagenette training benchmark.

C. Latency protocol

Latency is measured with TensorFlow Lite's XNNPACK CPU delegate on an AMD Ryzen 5 7600 running Windows build 26100. One interpreter thread is used, oneDNN graph

TABLE I
MEASURED MOBILENetV2 RESULTS. ACCURACY USES ALL 3,925 IMAGENETTE
VALIDATION IMAGES; LATENCY USES 500 IMAGES AFTER 50 WARM-UPS.

Metric	FP32	INT8
Model size (MiB)	13.34	3.81
Top-1 accuracy (%)	66.73	65.81
Top-5 accuracy (%)	90.75	89.81
Mean latency (ms)	8.53	6.46
Median latency (ms)	8.14	6.30
90th-percentile latency (ms)	9.88	7.29
Latency standard deviation (ms)	1.62	1.25
Throughput from mean (images/s)	117.2	154.8

optimizations are disabled, and preprocessing is excluded. After 50 untimed warm-up invocations, 500 deterministically shuffled validation inputs are timed individually with a monotonic nanosecond clock. The paper reports mean, median, 90th percentile, standard deviation, and throughput computed as the reciprocal of mean latency.

D. Serving and packaging

The FastAPI service loads models/mobilenet_v2_int8.tflite, validates upload size and image decoding, applies the evaluation resize, crop, and scaling procedure, quantizes according to tensor metadata, serializes interpreter access with a lock, and exposes health and metadata endpoints. Missing models return HTTP 503. The container runs as a non-root user with CPU and memory ceilings. These controls define deployment behavior; they are not treated as Raspberry Pi, Jetson, power, or container-overhead measurements.

III. RESULTS

Table I reports the complete validation and host-latency experiment. The FP32 model is 13.34 MiB and the INT8 model is 3.81 MiB, a 71.5% reduction or 3.50× compression. Mean latency decreases by 24.3%, from 8.53 to 6.46 ms, corresponding to a 1.32× speedup. The median and 90th-percentile values show the same direction.

The top-1 accuracy difference is 0.92 percentage points. FP32 top-1 accuracy has a Wilson 95% interval of 65.24–68.18%; INT8 has an interval of 64.31–67.28%. Top-5 accuracy changes by 0.94 percentage points, from 90.75% to 89.81%. These results indicate that the calibrated integer model closely tracks the floating-point model on this subset, while the overlapping intervals caution against interpreting the observed difference as a general ImageNet accuracy estimate.

A. Reproducibility and limitations

The result JSON stores all 500 raw latency observations, per-class accuracy counts, model hashes, runtime versions, dataset hash, calibration count and seed, preprocessing, and thread settings. The experiment can therefore be repeated or reanalyzed without transcribing values from the manuscript.

Three limits constrain interpretation. First, Imagenette covers only ten ImageNet classes, so the accuracy estimate does

not replace evaluation on the 50,000-image ImageNet validation set. Second, the latency result describes one x86-64 host and one thread. ARM CPUs and accelerators may produce different speedups. Third, preprocessing, HTTP transport, resident memory, container overhead, temperature, and power are outside the timed region. Claims about those quantities require separate, synchronized measurements on each target device.

IV. CONCLUSION

This study turns an edge-deployment concept into a reproducible conversion, evaluation, and serving artifact. Full-integer quantization compressed MobileNetV2 by 3.50× and improved single-thread host inference by 1.32× while changing top-1 accuracy by less than one percentage point on the full Imagenette validation split. The result supports the engineering value of calibrated INT8 conversion under a controlled host protocol.

The next empirical step is deployment on named ARM edge devices using the same model hashes and inputs. That study should isolate native inference from HTTP and container overhead, measure process memory in separate runs, and collect synchronized wall-power and temperature traces. The present repository supplies the model artifacts, dataset checksum, raw timings, and service needed to conduct that extension.

DATA AND CODE AVAILABILITY

Source code, model artifacts, test configuration, and machine-readable results are available at <https://github.com/imjbassi/efficient-ai-edge-deployment>. Imagenette is downloaded from the official fastai distribution and verified before extraction.

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